

Carlos De Manuel Vicente

Data Engineer

Valencia, Spain · open to relocation · carlosdmv7@gmail.com · (+34) 601 205 213
carlosdmv7.github.io/personal-portfolio · linkedin.com/in/carlos-de-manuel · github.com/carlosdmv7

I build and run production data pipelines at a global automotive manufacturer: near-real-time models over 800M+ rows in Snowflake, event-driven orchestration with Prefect, incremental loading and CI/CD, serving 20+ plants in North America, APAC and EMEA.

EXPERIENCE

Data & Analytics Engineer Jan 2025 – Present
RESRG Automotive · global automotive manufacturer Liria, Spain (hybrid)

- Built near-real-time analytical models for a warehouse management system over **800M+ rows** in Snowflake, refreshed by event-driven Prefect flows that fire when upstream Fivetran HVR replication lands, with the data engineering team.
- Cut stock data load times **~95%** with strict incremental loading in SQL, and automated supply-policy calculations and role-based inventory logic.
- Re-engineered the end-to-end refresh with CI/CD, cutting runtime **~98%** and compute cost **~90%**.
- Set one definition for **130+ operational KPIs** through a KPI-governance app built on dbt, with tests on every model and pipeline observability in Prometheus.
- Replaced manual Excel trackers across **20+ plants** (NA, APAC, EMEA) with self-serve data products in Streamlit and Power BI, from the executive summary **200+ people** open daily to inventory apps on the shop floor, across purchasing, finance, supply chain and production.

Data Scientist (internship) Nov 2023 – Jul 2024
Biomedical Data Science Lab, ITACA, UPV Valencia, Spain

- Built an end-to-end Python framework and preprocessing pipelines to evaluate healthcare ML models for fairness, robustness and explainability (XAI).

PROJECTS

Job Market Intelligence · [case study](#) · github.com/carlosdmv7/job-market-intelligence
Reads EU data-job ads every day and ranks them against a CV: six free APIs, LLM enrichment that went from **23 to 180** ads a day on the same free quota, dbt on MotherDuck, Prefect Cloud and Streamlit.

Spanish Housing Radar · [case study](#) · github.com/carlosdmv7/spanish-housing-radar
Finds under-priced flats in València by scoring every listing against its neighbourhood: weekly scraping, dbt with **162 data tests**, Prefect Cloud, Docker and Streamlit.

SKILLS

Platform & orchestration	AWS, Prefect, Airflow, GitHub Actions (CI/CD), Git, Azure DevOps, Docker
Warehouse & modelling	Snowflake, dbt, DuckDB, Kimball dimensional modelling
Languages	Python, SQL, Jinja, DAX
Quality & observability	pytest, Pydantic, ruff, mypy, Prometheus
BI & data apps	Power BI, Streamlit, Microsoft Fabric
AI & ML	Gemini and OpenAI APIs (incl. Azure OpenAI), LLM evals, scikit-learn, SHAP

EDUCATION

MSc, Data Analysis, Process Improvement & Decision Support Engineering · UPV, Valencia
BSc, Biomedical Engineering (ICT specialisation) · UPV, Valencia
Thesis graded 10/10 · distinction for academic performance (top 25%)

LANGUAGES

Spanish (native) · English (C1, Cambridge, 194)